# 3 Compartment Raise

03/10/2019

# Applicability I

*Large cross section of raise.

*Suitable for long raise.

*Raise at steep angle, where broken rock can flow by gravity.

*Raise has to be at fixed angle.

*Availability of timber is important to take care of high timber consumption.

*Raise of large cross section up to 24sqm is excavated at one time.

# Applicability II

*Raise with cross section more than 24 sqm in competent rock of high & medium strength, first raise up ward at reduced cross section than widened down ward to required size.

*Compartment raise is also a method of shaft excavation.

# Preparation I

*Area, centerline are marked at place, where raise has to be excavated.

*Center line for raise is fixed considering both sides, starting and point of puncher at upper level.

*First blast for raise is taken from floor of drive / cross cut.

*Next few blasts are taken by standing over accumulated muck pile.

*3 to 4  meter raising is possible with out erecting platform in raise.

# Preparation II

*Normally 1.2m round is drilled at face for blasting

*Drilling starts with 0.6m integrated drill steel rod.

# Compartment I

*Some time, number of compartment becomes there , one for broken rock, one for ladder, compressed air pipe, water pipe, third compartment for hoist, ventilation.

*After attending 4 to 5 meter height, raise is divided in two or three compartment depending up on cross section.

*Number of compartment are minimum two or more.

*One compartment is for holding broken rock is fitted with chute gate for loading broken rock in to mine car.

# Compartment II

*Chute compartment is of solid construction & is lined tightly with 50mm (2inch) planks (lagging) to with stand high side pressure of muck.

*Other compartment is filled with ladder, compressed air, water and ventilation duct pipe.

*Partition are built by fixing wooden logs, Sleepers by placing skin to skin.

*After every blast muck is drawn from bottom so that the level is maintained at the same height as that of man way compartment.

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# Compartment III

*Chute compartment is guarded by a railing to prevent accidental fall of personnel & larger piece of rock.

# Raising Process I

*Before blasting man, way compartment is covered by bulk head to divert blasted fragment to wards the muck compartment.

*Brocken rock from chute compartment is removed with help of chute gate located at bottom of raise directly in to car.

*Rock loaded cars are hauled by locomotive.

*Rock removal from bottom of raise has to be done at controlled rate, so that proper level of muck is maintained at  top.

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# Raising Process II

*Care has to be taken to see that plugging, arching does not take place in broken rock column, above chute gate.

*Only excess muck is removed by opening bottom damper so that after blasting, chute compartment should not over flow.

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# Raising Process III

*For long raise, a rebound niche is made every 10-30m in side wall of raise at particular side of raise.

*In hard rock, a raise face can be advanced up to 5m per month, with out lining & in weaker rock proportionally less.

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# Lining I

*When lining consists of timber set, every third set has to be inserted in shaft wall.

*When full timber lining is used, curb set support at least each platform of ladder compartment is done.

*A permanent lining is erected latter based on requirement.

*Good rock conditions does not require lining.

# Lining II

*For large cross section raise in in-competent rock, permanent lining is used.

*Raising shaft of small diameter, up to 4m, is done at full diameter with temporary lining in-competent rock.

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# Platform I

*Working platform in raise are made with timber planks strong enough to with stand load of piled on muck.

*There are two entrance with hatches at platform, one in ladder & one in hoist compartment.

*This platform is supported by steel guarders or wooden beams with strength adequate to support wooden platform, weight of muck, equipment & workers.

*For load carrying steel beam, a safety factor of six is recommended.

# Platform II

*Safety platform are lighter construction, lined with 50mm (2inch) lagging laid on beam of adequate strength.

*Distance between working & safety platform can not be less than 1.5m or more than 3m.

# Blasting I

*Blasting at raise face is done from level.

*Small dia blast holes are charged with indirect initiation.

*Blast hole direction is kept at such angle, so that most of waste rock falls in to chute compartment.

*Waste rock from chute compartment is drawn to make space for working at raise face.

*Normally, fan cut pattern is followed to reduce impact, so that possibility of damage is minimum.

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# Survey I

*Accuracy of survey of compartmental raise is low.

*There is no line of sight at face for survey.

*Raise can only be surveyed through man way.

*Survey stations has to come on compartment wall, which is not stable.

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# Ventilation I

*Ventilation in two compartment rise is done by placing ventilation duct in ladder way compartment.

*Brattice is installed from bottom of ladder compartment up to safety platform placed below working platform.

*For ventilating three compartment raise, one raise compartment is used as ventilation return, with help of exhaust fan.

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# Ventilation I

*For ventilation, a hole 200 to 300mm in diameter is drilled at center of raise.

*Hole is cased with steel pipe & projects above possible water level at upper level.

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# Advantages I

*Raising process is safe, less human exposure.

*It is possible to excavate long raise.

*Raise with wider cross section is possible to excavate.

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# Disadvantages I

*It is very difficult to repair damage / broken compartment.

*At end of raising, compartment, partition etc are to be removed before using raise.

*Draw of muck has to be strictly controlled to have proper platform on top for working.

*Possibility of accidental settling of broken rock can cause accident.

*It is difficult to remove plugging / arching in side compartment.

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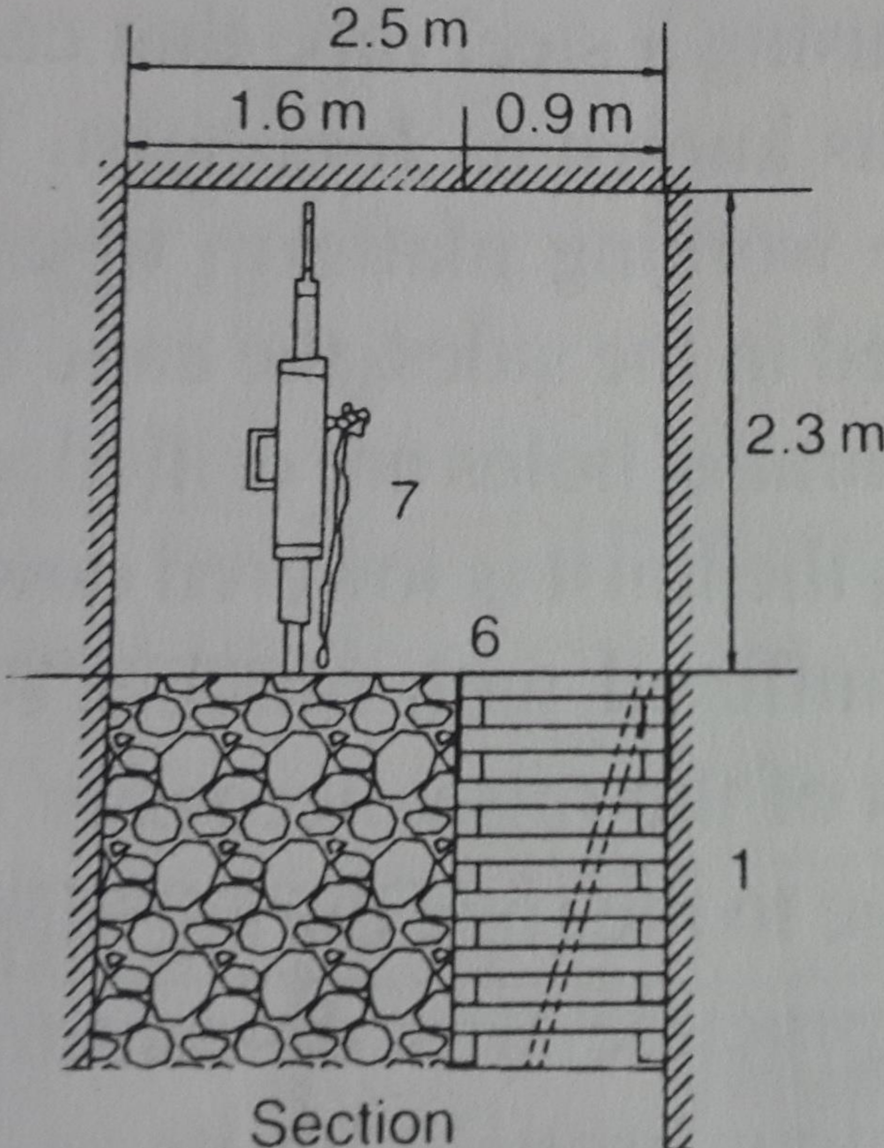
# Disadvantages II

*High timber consumption.

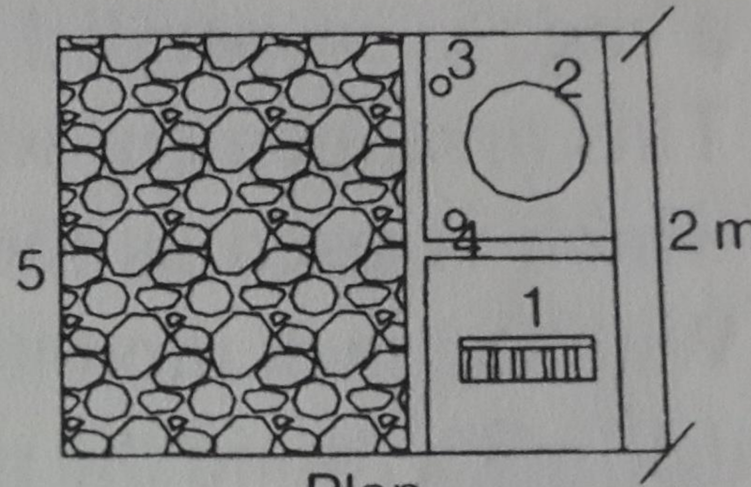
* Slow raise progress.

*Work is labour intensive.

# Compartmental Raise I




Section

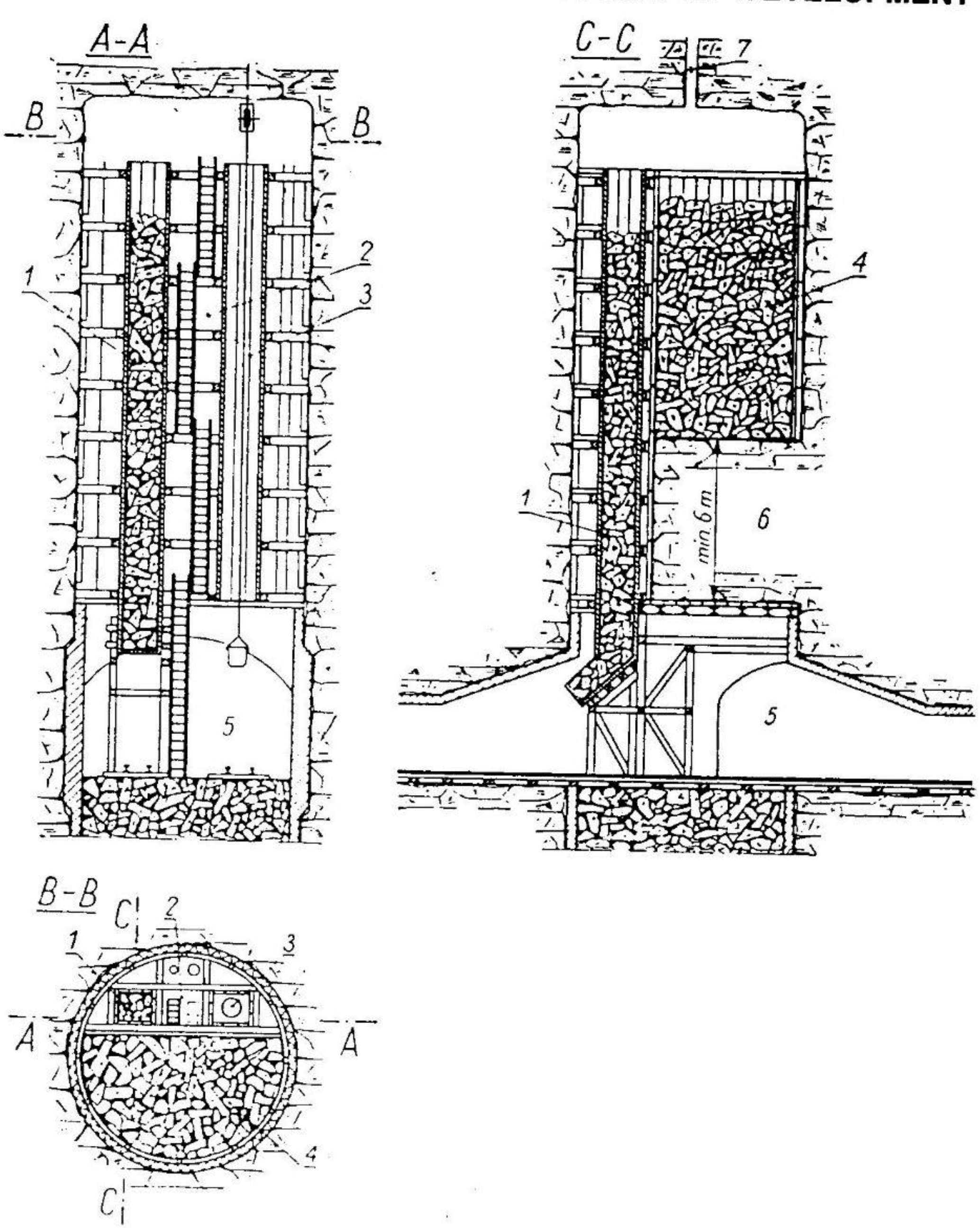



Plan

1 – Ladder
2 – Ventilation duct
3 – Compressed air pipe
4 – Water pipe
5 – Broken muck
6 – Partition timber
7 – Stoper drill

(b) Raising by compartment

# Compartment Raising




Raising method with full cross section, with temporary & latter erected permanent lining.

1- chute compartment

2-ladder compartment

3- bucket compartment

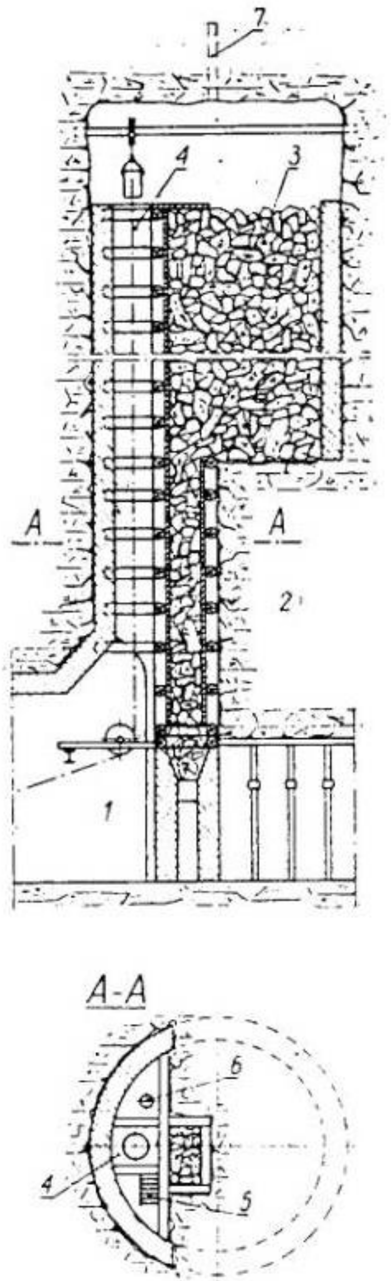
4- storage of muck

5- lower shaft station

6- protective self

7- ventilation hole

# Compartment Raising




Raising method with full cross section & with immediate permanent lining.

1- Lower Level

2- protective rock shelf

3- storage of muck

4- bucket compartment

5- ladder compartment

6- tubing compartment

7- drill hole for ventilation

# Reference I

*SME Mining Engineering Hand Book, 2nd Edition, Volume 2, Page 1621 to 1623.